

F729 — V_{us} blind: pin the FORCED Pochhammer arguments so Elie reproduces the Cabibbo angle forward, without ever using 0.2245 as input. The key that makes this a real test, not a fit: the mixing is the OFF-DIAGONAL overlap of the SAME discrete-series modes whose diagonal NORMS just gave the Derived $m_s/m_d = 20$. Those modes are read through the Faraut–Korányi generalized Pochhammer $(\nu)_\lambda$ at $\nu = N_c = 3$ (the Wallach threshold, forced) with degrees $\{1, 3, 5\}$ (d, s, b — from Q^5 cohomology, forced, Elie-blind T1929). The diagonal Pochhammers gave the masses ($m_s/m_d = (\nu)_3/(\nu)_1 = (N_c+1)(N_c+2) = 20$, banked K993); the $d \leftrightarrow s$ off-diagonal overlap of the same two modes gives V_{us} — the arguments are already pinned by the mass result, so V_{us} is forced, not free. Elie computes the normalized overlap forward; the observed 0.2245 stays walled off behind the comparison line (the tau-71 bar). If it lands blind, V_{us} promotes to Derived and the colored sector has its first forward mixing prediction — the credibility Casey flagged. This note pins the arguments; Elie runs the blind reproduction; the falsifiable payoff (V_{cb} , m_c/m_u , neutrino Δm^2) follows only if V_{us} lands clean.

Lyra, Wed 2026-07-29. V_{us} pin, corpus-first. The discipline: because the SAME modes give both m_s/m_d (diagonal) and V_{us} (off-diagonal), and m_s/m_d already fell out Derived and validated the addresses, V_{us} is a genuine forward test — no new freedom, no room to calibrate. A colored quark angle that falls out is a prediction; one that has to be aimed at is a red flag (F728).

The forced arguments (all pinned, all target-innocent, all validated by $m_s/m_d = 20$)

ingredient	value	forced by	validated?
the object	FK generalized Pochhammer $(\nu)_\lambda$	the Bergman machinery on D_{IV}^5 (in-corpus, F506/K671)	yes — gave $m_s/m_d = 20$
ν	$N_c = 3$	the Wallach threshold $k_{\min}$ (forced)	yes (same as the mass)
degrees λ	$\{1, 3, 5\} = d, s, b$	Q^5 cohomology (Elie-blind T1929)	yes (down ladder 1:20:840)
overlap form	$N(w)^{\{n_C/2\}}$	the type-IV norm / Bergman kernel (F87b/F184, forced)	— (the mixing form)

The pin (the one line Elie needs): V_{us} is the normalized off-diagonal overlap of the d (degree 1) and s (degree 3) modes, read through the same $(\nu=N_c)_\lambda$ Pochhammer that gave their norms:

$$V_{us} = \frac{\langle \psi_1 | \psi_3 \rangle}{\sqrt{\langle \psi_1 | \psi_1 \rangle \langle \psi_3 | \psi_3 \rangle}}, \quad \|\psi_\lambda\|^2 \propto (N_c)_\lambda, \quad \nu = N_c = 3, \quad \lambda \in \{1, 3\}.$$

The diagonal norms are $(N_c)_1 = 3$ and $(N_c)_3 = 60$ (giving the 20). The off-diagonal $\langle \psi_1 | \psi_3 \rangle$ is the *same* Bergman-kernel overlap evaluated between the two degrees — no new input: the addresses are fixed by the mass result, so the angle is determined. That is what makes this forced.

The blind protocol (Elie) — the tau-71 bar

1. Compute V_{us} = the normalized $d \leftrightarrow s$ Pochhammer overlap forward, from $\{N_c=3, \text{degrees } 1,3, \text{the overlap form}\}$, with the observed 0.2245 walled off.
2. Report the number. THEN compare.
3. Bar: it must fall out (like $m_s/m_d = 20$ did), not be tuned. Two corpus candidates to check the blind result against *after* (NOT to calibrate to): $1/\sqrt{\text{rank}^2 \cdot n_C} = 1/\sqrt{20} = 0.2236$ (0.4%), and $2/\sqrt{79} = \sqrt{(\text{rank}^2/(\text{rank}^4 \cdot n_C - 1))} = 0.2250$ (0.2%, the quantized-address form $N(w_\mu)^{n_C}=4/79$, Elie 4235). The blind overlap decides which — or neither. If the forward number matches without calibration $\rightarrow V_{us}$ Derived. If it only matches by tuning a coefficient $\rightarrow$ stays Tier-2/Identified, said plainly (the 71-failed discipline).

Why this is the make-or-break (Casey's frame)

- A calibrated V fails the bar — exactly as the tau's 71 failed (matched, not canonical). A colored mixing angle that has to be aimed at is a red flag, not a win.
- A forward V_{us} that lands is worth more than any postdiction — it's the first *prediction* on the colored sector, and it validates the whole $\mathbb{C}^3$ SVD address structure a second time (m_s/m_d was the first).
- The falsifiable payoff (only if V_{us} lands): the same pinned addresses then predict V_{cb} , m_c/m_u , and the neutrino Δm^2 — quantities we have never touched — forward and blind. That is the colored sector's credibility: not fitting the CKM we know, but predicting the entries we haven't used.

Tier / handoffs

- @Elie — run the blind $d \leftrightarrow s$ Pochhammer overlap on $\mathbb{C}^3$ (source the FK generalized Pochhammer + the overlap integral from FK Ch VI, do not reconstruct): $V_{us} = \langle \psi_1 | \psi_3 \rangle / \sqrt{((N_c)_1(N_c)_3)}$ with $\nu=N_c=3$, observed value walled off. Report the number, not a verdict. The addresses are the ones that gave $m_s/m_d=20$ — no re-pinning, no new freedom.
- @Keeper — rule V_{us} against the blind bar the instant it lands: fell-out-forward $\rightarrow$ Derived (promoting from the IDENTIFIED $1/\sqrt{20}$ lead); tuned $\rightarrow$ Tier-2/Identified, stated. Hold the over-fit line (the tau-71 precedent).
- @Cal — audit that the V_{us} overlap uses NO input beyond the mass-validated addresses ($\nu=N_c$, degrees $\{1,3\}$) — the one way this leaks is a free coefficient smuggled into the off-diagonal. Confirm the arguments are the same ones m_s/m_d rode.
- @Grace — the two candidate forms ($1/\sqrt{20}$, $2/\sqrt{79}$) are the after-comparison, walled off; pin the overlap-form $N(w)^{n_C/2}$ to F87b/F184 (the forced Bergman overlap), not a fitted power.
- @Casey — V_{us} is the honest test, and it's set up so we can't cheat: the mixing angle between the down and strange quarks is the *off-diagonal* of the very same little matrix whose *diagonal* just gave us $m_s/m_d = 20$ (which fell out, unaimed). So the angle is already determined — Elie just reads it off, with the measured Cabibbo value hidden behind a wall. If the number that comes out matches (near 0.2245) without anyone tuning it, that's a real prediction on the colored sector — the first one — and it means the addresses are pinned right twice over. If it only matches by fiddling a coefficient, we say Tier-2 and move on, exactly as we did when the tau's 71 turned out to be imported. And the prize, if it lands, is the part we've never touched: the same addresses would then predict the charm/up ratio, the V_{cb} angle, and even a neutrino mass-splitting — forward, blind, falsifiable. That's where the colored sector earns its credibility: predicting what we didn't use.

Notes only; no toy/theorem claimed (Elie owns the blind numeric). F729 (V_{us} blind pin): the Cabibbo angle is the OFF-DIAGONAL overlap of the SAME d,s modes whose diagonal norms gave $m_s/m_d=20$ (Derived K993) — so the forced Pochhammer arguments are already pinned: FK generalized Pochhammer (ν) $_\lambda$, $\nu=N_c=3$ (Wallach threshold), degrees $\{1,3\}$ (Q^5 cohomology), overlap form $N(w)^{n_C/2}$ (F87b). $V_{us} = \langle \psi_1 | \psi_3 \rangle / \sqrt{((N_c)_1(N_c)_3)}$, no new freedom. Elie computes forward, 0.2245 walled off (tau-71 bar). After-compare candidates (NOT calibration): $1/\sqrt{20}=0.2236$, $2/\sqrt{79}=0.2250$. Fell-out $\rightarrow$ Derived; tuned $\rightarrow$ Tier-2. Payoff if it lands: predict V_{cb} , m_c/m_u , neutrino Δm^2 forward/blind. Credibility = predicting what we didn't use. —

Lyra